when an attacker can send a URL to a webserver that causes it to serve a file that was intended to be private. For example, an attacker might ask for the file `/public/../../../etc/secrets/db-password`. Such vulnerabilities can reveal Kubernetes secrets to attackers.

11.5.2 Key and secret management services

An alternative to Kubernetes secrets is to use a dedicated service to provide credentials to your application. Secrets management services store credentials in an encrypted database and make them available to services over HTTPS or a similar secure protocol. Typically, the client needs an initial credential to access the service, such as an API key or client certificate, which can be made available via Kubernetes secrets or a similar mechanism. All other secrets are then retrieved from the secrets management service. Although this may sound no more secure than using Kubernetes secrets directly, it has several advantages:

- The storage of the secrets is encrypted by default, providing better protection of secret data at rest.
- The secret management service can automatically generate and update secrets regularly. For example, Hashicorp Vault (https://www.vaultproject.io) can automatically create short-lived database users on the fly, providing a temporary username and password. After a configurable period, Vault will delete the account again. This can be useful to allow daily